

8

Market structure and pricing

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Objectives

- 1 To define and explain the meaning of markets.
- 2 To explain the concept of market structure and its significance.
- 3 To describe the characteristics of the different types of market.
- 4 To examine the relationships between structure, conduct and performance.
- 5 To explain the equilibrium conditions for different types of market in terms of price and output, both in graphical and algebraic terms.
- 6 To explain the types and significance of entry and exit barriers.
- 7 To give examples of different industries where different market conditions exist, explaining their prevalence.
- 8 To examine some welfare implications regarding different forms of market.
- 9 To emphasize the importance of oligopolistic markets, and examine the particular problems relating to their analysis.

8.1 Introduction

In order to maximize profits or shareholder wealth, managers must use the information that they have relating to demand and costs in order to determine strategy regarding price and output, and other variables. However, managers

must also be aware of the type of market structure in which they operate, since this has important implications for strategy; this applies both to short-run decision-making and to long-run decisions on changing capacity or entering new markets.

It is useful to start by explaining the characteristics of markets and different types of market structure, with a general examination of the relationships between structure, conduct and performance. The four main types of market structure are then discussed and analysed in terms of their strategic implications.

8.1.1 Characteristics of markets

A **market** can be defined as **a group of economic agents, usually firms and individuals, who interact with each other in a buyer-seller relationship**. This is fine as a general definition but it lacks practical applicability in defining a specific market. How for example can we define the market for Coca-Cola? We can refer to a cola market, a carbonated drinks market, a soft drinks market, or a beverage market; all of these have different boundaries in terms of the number and types of product involved. Of course, Coca-Cola operates in all of these markets, but which definition is relevant? One might respond to this problem that this is purely a matter of semantics and is not a significant issue. This, however, is not the case. In 1986 the Federal Trade Commission (FTC) in the United States tried to block the merger between Coca-Cola and Dr Pepper, and the judgement in this case, as in many anti-trust cases, hinged on the definition of the relevant market.¹ In another case, in 1984, the FTC had to determine whether General Foods' Maxwell House brand of coffee had engaged in unfair competition with Proctor and Gamble's Folgers brand in certain US cities.² Again the FTC had to determine what constituted the relevant coffee market, this time in terms of geographical boundaries.

It can be seen from these examples that markets can be defined in terms of both product characteristics and geographical characteristics. In the first case the cross-elasticity of demand is an important measure in terms of indicating the degree of substitution between products. Thus in the Coca-Cola case it was determined that the relevant market was carbonated soft drinks. In the second case it was determined that the relevant market involved the whole of the United States, not just certain local areas, since the two companies competed nationwide. Of course, this raises an issue regarding global markets; many firms compete internationally, in which case the relevant market extends across national boundaries, even though this may pose problems for anti-trust authorities.

8.1.2 Types of market structure

Economists usually classify market structures into four main types: **perfect competition**, **monopoly**, **monopolistic competition** and **oligopoly**. These types of market structure are different according to the following characteristics: number of sellers, type of product, barriers to entry, power to affect

Table 8.1. Characteristics of different markets

Market structure	Number of sellers	Type of product	Barriers to entry	Power to affect price	Non-price competition
Perfect competition	Many	Standardized	None	None	None
Monopolistic competition	Many	Differentiated	Low	Low	Advertising and product differentiation
Oligopoly	Few	Standardized or differentiated	High	Medium	Heavy advertising and product differentiation
Monopoly	One	Single product	Very high	High	Advertising

price and the extent and type of non-price competition. Perfect competition (PC) is invariably used as a benchmark for comparison, in terms of price, output, profit and efficiency (of different types). It is therefore discussed before the other types of market, not because it is the norm in practice, but because it represents an ideal in certain respects. The conditions for PC are explained in more detail in the next section, but essentially it involves many buyers and sellers of a standardized product. An example would be farmers producing wheat (though strictly speaking there are different varieties of wheat). This type of market represents one extreme of the competition spectrum (maximum competition), while monopoly represents the other, where a single seller exists in a market. This again is rare in practice, but does tend to occur in public utilities like gas, water and electricity supply, particularly on a local basis. Monopolistic competition and oligopoly are intermediate cases, and are more frequently found in practice; the former involves many sellers of a differentiated product, for example restaurants, whilst the latter involves a few sellers dominating an industry, for example the car industry. The most important characteristics of these different markets are summarized in Table 8.1; in this table the degree of price competition decreases from one row to the next.

8.1.3 Relationships between structure, conduct and performance

Economists tend to relate **conduct** and **performance** in different markets to market structure. In this context, conduct refers to strategic behaviour, in particular in terms of price and output determination, but also in terms of advertising, product differentiation and practices regarding entry and exit barriers. This chapter focuses mainly on the aspects of price and output determination, while Chapters 9 and 11 examine other aspects of strategic behaviour. Performance refers to aspects of profit and efficiency. Traditionally the relationships between structure, conduct and performance have been viewed in terms of a one-way causation sequence: structure determines conduct which in turn determines performance. Structure itself has been seen as largely determined by technical factors, mainly the type of technology, but with government policy, infrastructure and geographical factors all being relevant. Currently the relationships between the different aspects are seen as being more complex, with various feedback mechanisms between them.

These will be seen as the chapter progresses and each type of market structure is examined in detail. The involvement of the government in terms of these relationships is examined in Chapter 12.

8.1.4 Methodology

For each type of market structure the necessary conditions for its existence are first explained. A static, partial equilibrium analysis then follows, usually assuming certainty or perfect information. This analysis is initially graphical, following the lines of introductory microeconomics texts with which the student may be familiar. The analysis then moves on to an algebraic approach involving optimization techniques. A knowledge of basic calculus is assumed at this point. Some of the limitations of these methods are then discussed, but the implications of these limitations are not developed until Chapter 9, when strategic behaviour in a dynamic environment under conditions of uncertainty is examined in more detail.

8.2 Perfect competition

It should be understood that this form of market structure is not necessarily 'perfect' in an economic sense, i.e. resulting in an optimal allocation of resources, although this may happen. It simply represents a situation where competition is at a maximum; it is therefore sometimes referred to as pure competition or atomistic competition.

8.2.1 Conditions

There are five main conditions for perfect competition to exist:

- 1 *Many buyers and sellers.* Each of these must buy or sell such a small proportion of the total market output that none is able to have any influence over the market price.
- 2 *Homogeneous product.* Each firm must be producing an identical product, for example premium unleaded petrol or skimmed milk.
- 3 *Free entry and exit from the market.* This means that there are no barriers to entry or exit that give incumbent firms an advantage over potential competitors who are considering entering the industry. These barriers, which can represent either demand or cost advantages, are explained in more detail in the next section.
- 4 *Perfect knowledge.* Both firms and consumers must possess all relevant market information regarding production and prices.
- 5 *Zero transportation costs.* This means that it does not cost anything for firms to bring products to the market or for consumers to go to the market.

Under the above conditions, firms in the market will be **price-takers**; there will be one, and only one, market price, meaning that the product will sell at the same price in all locations. Most treatments of perfect competition

concentrate on the first four conditions, but the last one is relevant, as will be seen in the following example. It is not uncommon for motorists driving along a busy road to see two or three petrol stations close to each other, at a major crossroads for example, with these firms selling petrol at somewhat different prices. There are many buyers and sellers of petrol, the different prices can be for the same homogeneous product like premium unleaded petrol, barriers to entry are not relevant in the short run and consumers can see the different prices being charged by the different petrol stations. Yet these price differentials may persist because motorists may not be willing to cross the road to obtain a cheaper price and then cross the road again to continue in their intended direction. This is quite rational if the opportunity cost of the time taken to cross and recross the road exceeds the savings on the cost of the petrol. Thus, in practice, transportation costs can be important in explaining price differentials which could not be explained in terms of the first four conditions.

It is clear that these conditions are rarely achieved in reality; however, this does not negate the usefulness of the analysis of perfect competition. Once again it is necessary to remember that a theory should be judged not on the basis of the realism of its assumptions but on its ability to explain and predict. The conditions are approximately approached in some markets, for example agricultural products and stock markets. Furthermore, the analysis and conclusions in terms of conduct and performance provide a useful benchmark for comparing other forms of market structure.

8.2.2 Demand and supply

In order to perform either a graphical or algebraic analysis we must consider the determination of demand and supply functions. It is important to distinguish between the individual firm and the industry as a whole; we will discuss the individual firm first.

a. The firm's demand function

We have just seen that under the conditions of PC the firm will be a price-taker. This means that each firm faces a perfectly elastic (horizontal) demand curve, at the level of the prevailing market price. The economic interpretation of this is that if the firm charges above the market price it will lose all its customers, who will then buy elsewhere, and that there is no point in charging below the market price, because the firm can sell all it wants, or can produce, at the existing price. The level of the prevailing market price is determined by demand and supply in the industry as a whole, as shown in Figure 8.1(i). The demand curve in this case represents both average revenue (AR) and marginal revenue (MR), since both of these are equal to the market price.

b. The firm's supply function

This refers to the quantities of a product that a firm is willing to put onto the market in a given time period at different prices. It can be seen that this

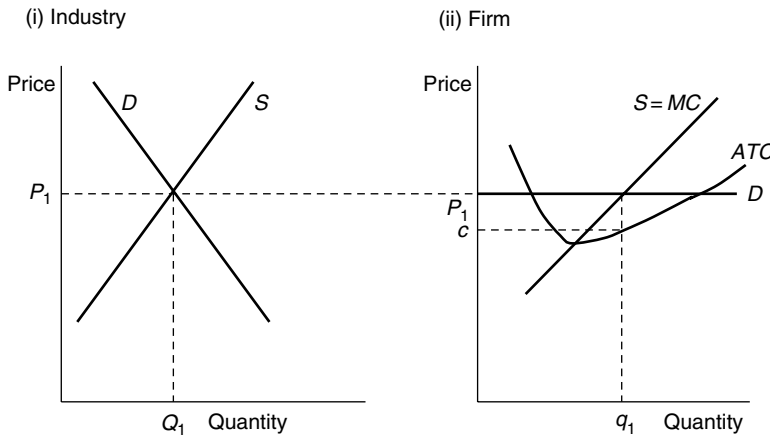


Figure 8.1. Short-run equilibrium in perfect competition.

definition corresponds in terms of its parameters to the definition of demand given in Chapter 3. In order to derive the firm's supply curve we must make the assumption that the firm wishes to maximize profit. In order to do this it must produce the output where marginal revenue equals marginal cost, as with any firm. Since the firm is a price-taker the marginal revenue is given by the price; thus the firm maximizes profit by producing the output where price equals marginal cost. It was seen in Chapter 6 that this curve was generally upward-sloping in the short run. Therefore, the firm will tend to produce more output as the market price increases, and its supply curve will be its marginal cost curve, as shown in Figure 8.1(ii). More precisely, the firm's short-run supply curve will be that part of its marginal cost curve that lies above the average variable cost (AVC) curve; if the price falls below the minimum level of AVC the firm will shut down production, since it will not be able to cover its variable costs let alone make any contribution to fixed costs.

c. The industry's demand function

As explained in Chapter 3, this can be viewed as the sum of all the individual consumers' demand functions; graphically, these are summed horizontally.

d. The industry's supply function

This is obtained by summing the supply functions of all the individual firms in the industry; this is again a horizontal sum when represented graphically. However, this is actually an oversimplification, since when the behaviour of firms is aggregated this has an effect on input prices, bidding them up.

8.2.3 Graphical analysis of equilibrium

We can now show the equilibrium situation for a firm and industry in perfect competition. This is represented as a static short-run equilibrium. The equilibrium

market price, P_1 , is determined by the demand and supply functions in the industry as a whole. The firms in the industry, as price-takers, then have to determine what output they will supply at that price. This output, q_1 , is where $P_1 = MC$, since this will maximize profit. It is important to distinguish between short-run and long-run equilibrium, as will be explained.

a. Short-run equilibrium

Whether the situation illustrated in Figure 8.1 is also a long-run equilibrium depends on the profit that is made. Since it is assumed that opportunity costs are included in the measurement of unit costs, as explained in Chapter 6, it can be deduced that if the market price is equal to average total cost (ATC), the firm will just make **normal profit**. This is defined as **the profit that a firm must make in order for it to remain in its current business**. If a firm cannot cover all its opportunity costs, meaning that $P < ATC$, then it should leave the industry in the long run since the owners of the business can use the resources more profitably elsewhere. If $P < AVC$ then the firm should shut down in the short run since it cannot even cover its variable costs, let alone make any contribution to fixed costs.

However, if $P > ATC$, the firm is making **abnormal** or **supernormal** profit. In Figure 8.1(ii) this is given by $(P_1 - c)q_1$. This means that the industry is more profitable than average and this will in turn attract new firms into the industry in the long run, since it is assumed that there are no entry or exit barriers. This long-run situation can now be considered.

b. Long-run equilibrium

It is possible in the long run for firms to enter or leave the industry; it should be recalled that this is not feasible in the short run, since it involves a change in the level of fixed factors employed. Existing firms can also change their scale and will maximize profit by producing where price equals long-run marginal cost ($P = LMC$). The presence of abnormal profit will always serve to attract new entrants into the industry, and this factor, combined with firms increasing their scale, will shift the industry supply curve to the right, in turn causing the market price to fall. The industry supply curve will shift from S_1 to S_2 in Figure 8.2(i), and the market price will fall from P_1 to P_2 . It is assumed here that there is no change in demand. Such changes are examined in a later section.

In order to aid the visual presentation, two separate diagrams are shown for the firm's situation. Figure 8.2(ii) shows the comparison of the two short-run equilibria, while Figure 8.2(iii) compares the new short-run equilibrium with the long-run equilibrium.

At the new equilibrium the price and long-run average cost (LAC) of the firm are equal, and the firm is now producing at the minimum level of both its new SAC curve (SAC_2) and its LAC curve at the output q_2 . This is shown in Figure 8.2(iii). At this point all the abnormal profit has been 'competed away'. There is, therefore, no further incentive for firms to enter the industry and the firm

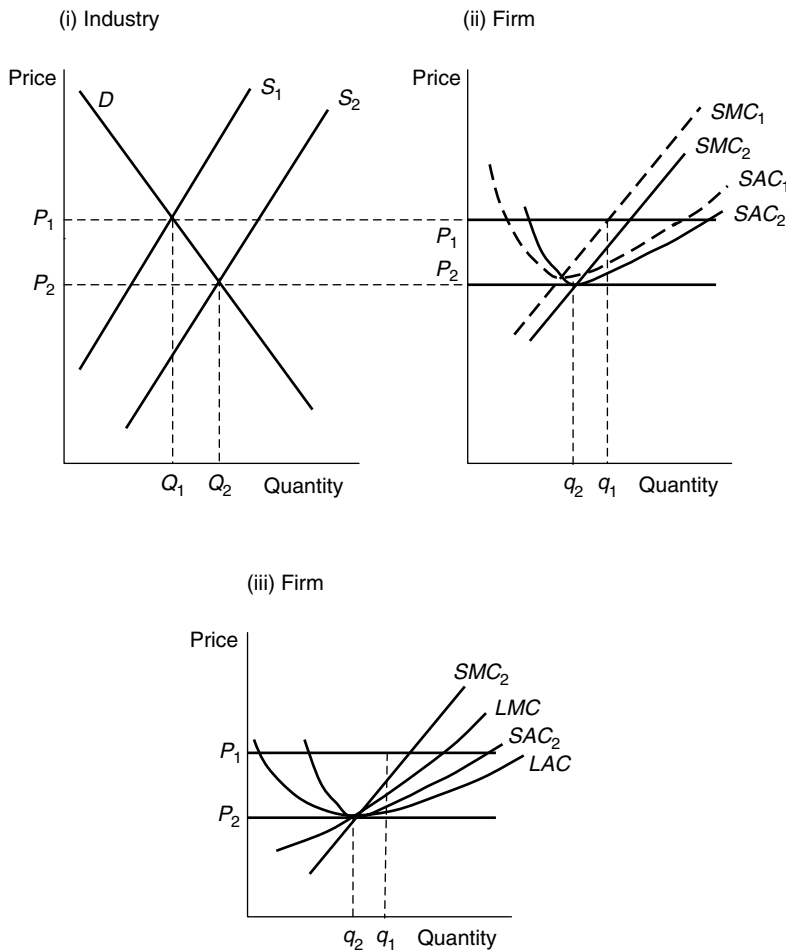


Figure 8.2. Long-run equilibrium in perfect competition.

and industry are in long-run equilibrium. If there were an excessive entry of new firms this would shift the industry supply curve even further to the right, the price would fall below P_2 , and firms would not even be able to make normal profit. Some firms would therefore leave the industry until the price rose back to P_2 .

It is helpful to remember in this graphical analysis that essentially a two-step procedure is involved:

- 1 Determining the **profit-maximizing output**. This means examining P and MC .
- 2 Determining the **size of the profit**. This means examining P and ATC .

In this long-run equilibrium the following conditions hold:

$$P = SMC = LMC = SAC = LAC = MR = AR \quad (8.1)$$

The significance of this is that both **productive** and **allocative efficiency** are achieved. Productive efficiency has already been explained in Chapter 6; it refers to the situation where the firm is **producing at the minimum level of average cost**. This is achieved in the long run and the firm is using the optimal scale with maximum efficiency. Note that in the long run, even though the firm is using a larger scale, it is producing less output than in the short-run situation, q_2 compared with q_1 . This is because it was using the scale inefficiently in the short run, producing too much output. Note also that this inefficiency resulted from the firm maximizing its profit, showing that profit maximization and efficiency are not equivalent. Allocative efficiency will be explained in the next section on monopoly, when the two types of market structure are compared.

8.2.4 Algebraic analysis of equilibrium

The graphical analysis above is useful and illustrates many insights into the nature of equilibrium. However, algebraic analysis tends to be a more powerful method when it comes to modelling the situation and making forecasts.

Consider the following situation. The market for milk in a certain town is perfectly competitive. The market demand is given by:

$$Q = 16 - 20P$$

Alternatively this can be written as:

$$P = 0.8 - 0.05Q \quad (8.2)$$

where P is the price of milk in £/litre and Q is the quantity of milk in the market in hundreds of thousands of litres per day. There are a hundred firms in the industry and each has the following marginal cost function:

$$P = 0.44 + 4q \quad (8.3)$$

where q is the quantity produced by each firm.

Firms also have fixed costs of £80 per day. We can now derive both the short- and long-term equilibria in this market; it will be assumed for simplicity that the firm is currently operating at its most efficient scale (with the equivalent of the marginal cost curve SMC_2 in Figure 8.2).

First we must obtain the market supply curve by summing all the firms' supply curves. Each firm's supply curve is given by its marginal cost curve. The equation for this curve can be turned around to give:

$$q = -0.11 + 0.25P$$

Therefore the industry supply curve is given by $Q = 100q = -11 + 25P$

This can be turned around to give:

$$P = 0.44 + 0.04Q \quad (8.4)$$

The equations (8.2) and (8.4) can be solved simultaneously to give:

$$0.36 = 0.09Q \text{ so } Q = 4, P = 0.6$$

Thus the short-run equilibrium price in the market is **£0.60/litre** and the quantity produced is **400,000 litres per day**.

Each firm will sell **4,000 litres per day** ($q = 0.04$), and its average cost is calculated by substituting this value into its *ATC* function. This function can be obtained by considering the firms' *TC* function. The *TC* function is obtained by integrating the *MC* function. This gives:

$$TC = a + 0.44q + 2q^2$$

where a represents the fixed costs of the firm.

Since *TC* is measured in hundreds of thousands of pounds (because of the units of q and Q), this gives:

$$TC = 0.0008 + 0.44q + 2q^2 \quad (8.5)$$

and

$$ATC = 0.0008/q + 0.44 + 2q \quad (8.6)$$

Therefore

$$ATC = 0.02 + 0.44 + 0.08 = 0.54$$

This means that each firm will make an economic profit given by $(0.6 - 0.54)0.04 = 0.0024 = \text{£}240 \text{ per day}$. In the long run this will attract new firms into the industry and drive the price down to the minimum level of long-run average cost. Since it has been assumed that the firm is already operating at the most efficient scale, we have to find the minimum level of average cost for this scale and this will also be the minimum level of long-run average cost. We therefore have to differentiate the *ATC* function and set the derivative equal to 0:

$$\frac{d(ATC)}{dq} = -0.0008q^{-2} + 2 = 0$$

This gives $q = 0.02$ (2,000 litres per day) and

$$ATC = 0.04 + 0.44 + 0.04 = 0.52.$$

Therefore the long-run equilibrium price is **£0.52 per litre** and the market output is given by the demand function (8.3):

$$0.52 = 0.8 - 0.05Q; Q = 5.6 \text{ or } \text{560,000 litres per day}.$$

8.2.5 Adjustment to changes in demand

Now that we have examined the short- and long-run equilibria for a perfectly competitive market we can consider how such a market adjusts to changes in

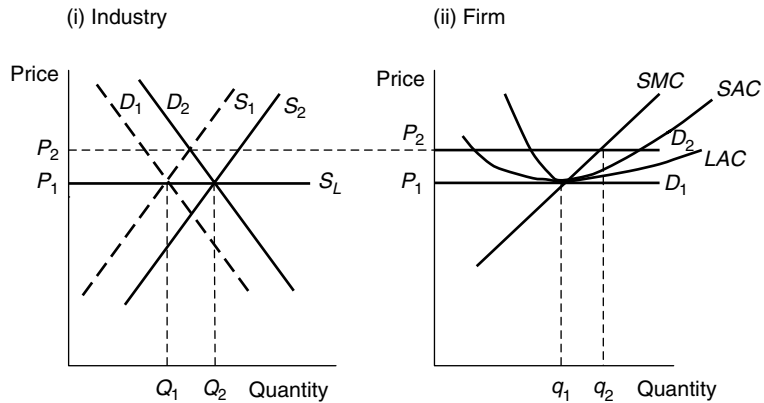


Figure 8.3. Adjustment under constant costs.

demand. The nature of the adjustment depends on the cost conditions under which the industry is operating. In this context the cost conditions refer to the existence of external economies and diseconomies of scale, which were defined in Chapter 6. Under **constant cost conditions** the firm experiences **no external economies or diseconomies** when the industry grows in size. Under **increasing cost conditions** the firm experiences **external diseconomies of scale**; these can be caused by the prices of inputs being driven up by the increased demand for them. Thus firms may have to pay higher raw material costs and wages as the industry expands, particularly if the industry has already reached a mature phase. This means that the firms will find both short- and long-run unit cost functions shifting upwards. Under **decreasing cost conditions** the firm experiences **external economies of scale**; in this case input prices are falling as the industry expands. This is most likely to happen in the initial growth phase of the industry, as the presence of other firms, particularly in the same locality, helps to provide an infrastructure of complementary firms, services and skilled labour. This has the effect of shifting the unit costs downwards. These different cost conditions are now examined graphically.

a. Constant costs

The effects of an increase in demand are shown in Figure 8.3. In the short run the increase in demand causes an increase in price (from P_1 to P_2), since firms face increasing marginal costs in producing more output (q_2 instead of q_1) and no new firms can enter the industry in this time frame. In the long run the supply curve shifts to S_2 as more firms enter the industry, the price falls back to P_1 at the minimum point of LAC , and the firms' demand curve falls from D_2 back to D_1 . Each firm in the industry ends up selling the same amount as originally (q_1), but with more firms the total industry output increases from Q_1 to Q_2 . The long-run supply curve in the industry is therefore horizontal, as shown by S_L .

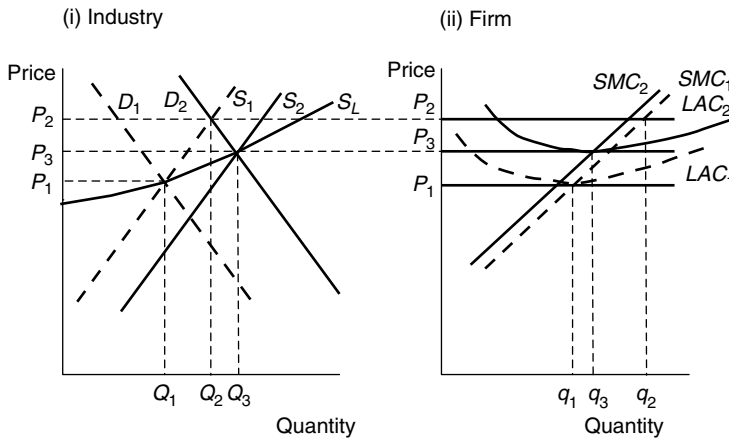


Figure 8.4. Adjustment under increasing costs.

b. Increasing costs

The effects of an increase in demand in this situation are shown in Figure 8.4. In the short run the increase in demand causes an increase in price (from P_1 to P_2), since firms have to move along their SMC_1 curve in producing more output (q_2 instead of q_1) and no new firms can enter the industry in this time frame. In the long run the supply curve shifts to S_2 as more firms enter the industry, and the price falls back to P_3 . This is higher than the original price because as more firms enter the industry the firms' cost curves shift upwards, so that the firms end up producing at the minimum point of LAC_2 . Each firm in the industry ends up selling the amount q_3 , which is more than their original output (q_1), but less than immediately after the change in demand (q_2). With more firms the total industry output increases from Q_1 to Q_3 . The long-run supply curve in the industry is therefore upward-sloping, as shown by S_L .

c. Decreasing costs

The effects of an increase in demand in this situation are shown in Figure 8.5. In the short run the increase in demand causes an increase in price (from P_1 to P_2), since firms again have to move along their SMC_1 curve in producing more output (q_2 instead of q_1) and no new firms can enter the industry in this time frame. In the long run the supply curve shifts to S_2 as more firms enter the industry, and the price falls back to P_3 . This is lower than the original price because as more firms enter the industry the firms' cost curves shift downwards, so that the firms end up producing at the minimum point of LAC_2 . Each firm in the industry ends up selling the amount q_3 , which is more than their original output (q_1), but less than immediately after the change in demand (q_2). With more firms the total industry output increases from Q_1 to Q_3 . The long-run supply curve in the industry is therefore downward-sloping, as shown by S_L .

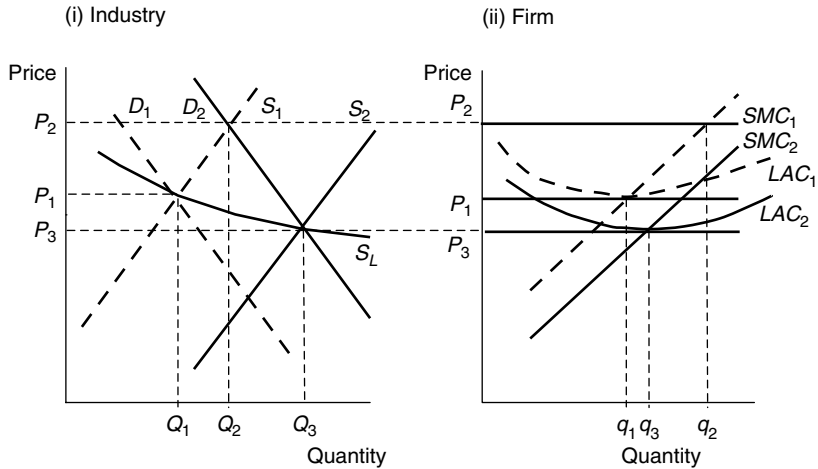


Figure 8.5. Adjustment under decreasing costs.

8.3 Monopoly

After an analysis of perfect competition it is usual to consider monopoly next. Again, this is not because this is a frequently found type of market structure, but because as an extreme form it provides a benchmark for comparison. As we shall see, it can also be claimed that many firms are in effect limited monopolies.

8.3.1 Conditions

Economists have defined monopoly in many different ways. Literally, it means a **single seller** in an industry. However, it is preferable to define a monopoly as being **a firm that has the power to earn supernormal profit in the long run**. This ability depends on two conditions:

- 1 *There must be a **lack of substitutes** for the product.* This means that any existing products are not very close in terms of their perceived functions and characteristics. Electricity is a good example.
- 2 *There must be **barriers to entry or exit**.* These are important in the long run in order to prevent firms entering the industry and competing away the supernormal profit. We now need to examine them in detail.

8.3.2 Barriers to entry and exit

These can be defined as factors that allow incumbent firms to earn supernormal profits in the long run by making it unprofitable for new firms to enter the industry. It is useful to distinguish between **structural** and **strategic**

barriers. Structural barriers, often referred to as natural barriers, occur because of **factors outside the firm's control**, mainly when an incumbent firm has natural cost or marketing advantages, or is aided by government regulations. Strategic barriers occur when an incumbent firm **deliberately deters entry**, using various restrictive practices, some of which may be illegal.

a. **Structural barriers**

There are six main types of structural barrier.

1. Control of essential resources. This often occurs for geographical reasons, because of the concentration of such resources in certain areas. For example, oil, gas and diamonds are only found in limited supplies and locations; certain areas are very advantageous for the production of quality wines. These resources may also include the expertise of resource owners, who may be skilled surgeons or scientists. Patent ownership is relevant here, although this also relates to the sixth barrier described later.

2. Economies of scale and scope. If these are significant it means that the minimum efficient scale (MES) will be large in the industry, as seen in Chapter 6. In order for new firms to compete in terms of cost they will have to enter on a large scale and this may be a problem if the MES is large compared to market demand. The problem is that the entry of another large firm in the industry would cause a substantial reduction in the market price, thus reducing profitability. Of course the industry would no longer be a monopoly in this situation, but the problem remains that market demand may not be sufficient to support two firms in the industry if the MES is large and the cost structure is high. This situation applies in particular to public utilities, like gas, water and electricity supply; such industries are, therefore, sometimes referred to as natural monopolies.

3. Marketing advantages of incumbents. Brand awareness and image are very important in many industries, with consumers being unwilling to buy unknown brands. This applies more to oligopolistic industries, like electrical and electronic appliances, banking and other financial services, but may also relate to industries like telecommunications. Such brand awareness can take time and a heavy cost to develop.

4. Financial barriers. New firms without a track record find it more difficult and more costly to raise money, because of the greater risk they impose on the lender. This disadvantage does not apply to the same extent to existing firms who are entering a new industry, although even here there are usually additional risks.

5. Information costs. In order for a new firm to enter an industry, or an existing firm to enter a new industry, much market research needs to be

carried out to investigate the potential profitability of such entry. This again imposes a cost.

6. Government regulations. Patent laws have already been mentioned as a reason for blocking the entry of new firms. They are particularly important in an industry like the pharmaceutical industry, where it takes a long time to get approval to use a product after it has originally been developed. Many governments also deliberately create monopolies through a licensing system. Thus public utilities and postal services in many countries operate as legally protected monopolies, with only one firm being allowed to supply the market.

Some of the above barriers also serve as **barriers to exit**; such barriers are in the form of sunk costs. From Chapter 6 it should be remembered that these are costs that are not affected by a particular decision; this means that once the decision to enter an industry has been made, these **costs cannot be recovered by the entrant** should it leave the industry. These costs include advertising costs to create brand awareness, market research costs, loss on the resale of assets, redundancy payments that have to be paid to workers, and so on. The existence of such costs increases the risk of entering a new industry.

b. Strategic barriers

It may be profitable for monopolists to employ strategies that deter entry if such strategies change the expectations of the potential entrants regarding the nature of the market after they have entered. The following strategies to deter entry may be used.

1. Limit pricing. This refers to the practice where an incumbent tries to discourage entry by **charging a low price before any new firm enters**. However, it has been shown using game theory that this strategy should only work if the potential entrant does not know the cost structure of the incumbent. Otherwise the firm will enter anyway, believing that any price reduction by the incumbent is only temporary; this is because it is rational for the incumbent to raise price after entry, since this would maximize its profit. On the other hand, if the potential entrant does not know the incumbent's cost structure, a low price may fool it into believing that the incumbent has low costs and can profitably maintain the low price³. In this case it would not be profitable to enter. The behaviour of incumbent and potential entrant may change over time as the latter may gain more accurate knowledge of the incumbent's cost structure and therefore may be more inclined to enter. Airlines, in particular Continental, have practised limit pricing to discourage competitors on particular routes. Note that the monopoly is limited in this case to that route.

2. Predatory pricing. This refers to the practice where an incumbent tries to encourage exit, meaning drive firms out of the industry, by **charging a low price after any new firms enter**. Again it has been shown that this strategy should only work if the new entrant does not know the cost structure of the

incumbent. The reasoning is basically the same as with limit pricing, and has been illustrated by what is known as 'the chain-store paradox'.⁴ In the United States and some other countries predatory pricing is prohibited by law.

3. Excess capacity. Most firms typically operate at about 80 per cent of full capacity. Firms may have extra capacity for a number of reasons, either deliberate or accidental, but excess capacity can serve as a credible threat to potential entrants. This is because it is easy for incumbents to expand output with little extra cost, thus forcing down the market price and post-entry profits; if these profits are less than the sunk costs of entry, the entrant will be deterred from entering the market. This would apply even if the potential entrant had full knowledge of the incumbent's cost structure.

4. Heavy advertising. This forces the potential entrant to respond by itself spending more on advertising, which has the effect of increasing its fixed costs, thus increasing the minimum efficient scale in the industry. It also adds to the marketing advantages of the incumbent, as discussed earlier.

These practices will not be possible if the market is **contestable**. The concept of contestable markets was developed by Baumol, Panzar and Willig⁵ and is examined in more detail in the next chapter. The following conditions are necessary for a market to be contestable:

- 1 There are an unlimited number of potential firms that can produce a homogeneous product, with identical technology.
- 2 Consumers respond quickly to price changes.
- 3 Incumbent firms cannot respond quickly to entry by reducing price.
- 4 Entry into the market does not involve any sunk costs.

Under such conditions a monopolist cannot raise price above the level of a perfectly competitive market. The result would be that a firm could enter on a hit-and-run basis, by undercutting the incumbent, and exiting quickly if the incumbent retaliates. Railway operating companies and airlines are generally quoted as examples of industries where such conditions may apply, since firms are supposed to be able to sell the necessary rolling stock or aircraft on secondary markets with no loss.

Unfortunately for the theory there appears to be little empirical evidence supporting it, at least in terms of the airline industry,⁶ where it was supposed to apply. This appears to be because it takes longer for firms to enter or leave the industry than it takes to adjust prices. Empirical evidence suggests that entry and exit barriers are often high, causing considerable differences in profitability, both between firms in the same industry and between different industries. Although entry is frequent, both by new firms and by existing firms in other markets, exit is also frequent, and entry does not appear to be that responsive to differences in profitability, or to affect profitability significantly.⁷ The most that can be claimed regarding contestability is that some

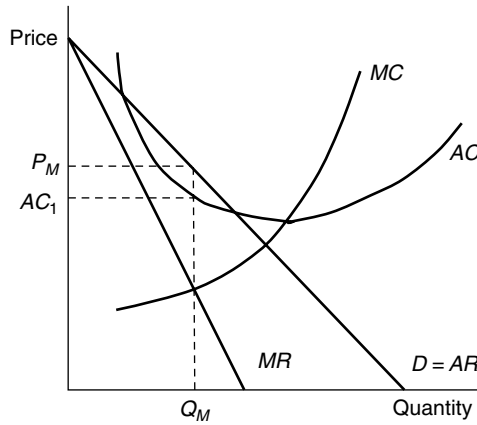


Figure 8.6. Equilibrium in monopoly.

markets are partially contestable, in that monopolists may reduce prices somewhat under the threat of potential competition.

This completes the discussion of entry barriers for the moment. Further aspects of this topic will be covered in the next chapter, particularly those relating to game theory and making credible threats, for example threats to enter a market or reduce price.

8.3.3 Graphical analysis of equilibrium

In the case of monopoly the firm and the industry are one and the same thing, and therefore only one graph needs to be drawn. The graph in Figure 8.6 shows a short-run equilibrium situation, but the same will apply in the long run since barriers to entry will prevent new firms from entering. The only difference is that the relevant cost curves would be long-run, as opposed to short-run, cost curves (LMC and LAC instead of SMC and SAC). It should be noted that there is no supply curve in this case. This is because a supply curve shows the quantities that producers will put onto the market at different prices, thus assuming that firms are price-takers. A monopoly on the other hand is a price-maker.

This also has implications on the demand side. In perfect competition, price, average revenue and marginal revenue are identical. Thus if the price is £0.52 per litre this will also be the average revenue and marginal revenue for the firm. This is not the case in monopoly, or indeed in any situation where the demand curve is less than perfectly elastic, in other words is downward-sloping. This is because in order to sell more of the product the firm must reduce its price not just on the additional products sold but also on all the other units (unless it is able to practise price discrimination, explained in Chapter 10). This means that marginal revenue will always be less than average revenue. It was shown in Chapter 3 that for a linear demand curve the marginal revenue curve

will have twice its slope, and this was illustrated in Figure 3.9. The same principles apply to the situation shown in Figure 8.6.

The profit-maximizing output is given by Q_M , where $MC = MR$. If the monopolist produces less than this output it can increase profit by producing more because it will be adding more to revenue than to costs. Similarly, if the monopolist produces more than Q_M it will be reducing its profit, since it will be adding more to costs than to revenue. The price that this output will fetch on the market is determined by the demand curve; this is the price P_M . Readers should check that they understand the nature of this profit-maximizing position, since it is of fundamental importance in much optimization behaviour.

The profit of the monopolist can also be seen from the graph; the profit per unit is given by $P_M - AC_1$, therefore total profit is given by the area of the rectangle $(P_M - AC_1)Q_M$. Again, as with perfect competition, it should be noted that there is a two-stage procedure involved in the analysis, the first stage examining profit maximization and MC and MR , with the second stage examining the size of the profit and AC and AR .

8.3.4 Algebraic analysis of equilibrium

Let us now assume that the milk market described and analysed in subsection 8.2.4 is supplied by a monopoly rather than a perfectly competitive market. In this case the marginal cost curve will be linear rather than quadratic as portrayed in the previous section, but otherwise the analysis is similar. We therefore have the following functions describing the market:

$$\text{Demand: } Q = 16 - 20P \quad (8.7)$$

$$\text{Marginal cost: } MC = 0.44 + 0.04Q \quad (8.8)$$

$$\text{Total cost: } TC = 0.08 + 0.44Q + 0.02Q^2 \quad (8.9)$$

We are assuming that the fixed costs of the monopolist are the sum of the fixed costs of all the competitive firms. The units are the same as previously. We can now determine the equilibrium of the monopolist by deriving the MR function and setting it equal to the MC function in order to solve for Q . This value can then be substituted into the demand equation to solve for P . To obtain MR we first need to obtain a revenue function in terms of output by multiplying P (given by the demand function) by Q . These steps are now shown in the appropriate sequence:

- 1 $P = 0.8 - 0.05Q$
- 2 $R = PQ = (0.8 - 0.05Q)Q = 0.8Q - 0.05Q^2$
- 3 $MR = 0.8 - 0.1Q$
- 4 $MC = MR$
 $0.44 + 0.04Q = 0.8 - 0.1Q$
 $0.14Q = 0.36$

$$Q = 2.57143 \text{ or } \mathbf{257,143} \text{ litres per day}$$

$$P = 0.8 - 0.05(2.571) = \mathbf{£0.67}$$

The price is higher and the output lower than in perfect competition and this comparison will be developed further in subsection 8.3.6. Profit can also be calculated by computing the total costs and revenues. Total revenue is £172,286, total cost is £134,244, and this gives a total profit of £37,942 per day. This compares with total profit in the industry of £24,000 per day under perfect competition (in the short run).

8.3.5 Pricing and price elasticity of demand

There are some important relationships between the optimal price (assuming a profit-maximizing objective) and price elasticity of demand (PED). These relationships do not just relate to monopolists but to any firm that has some element of control over price, or in other words faces a downward-sloping demand curve. The simplest of these relationships involves profit margin.

a. Profit margin

Profit margin is defined as **the difference between the price and the marginal cost, expressed as a percentage of the price**. It can thus be written as:

$$M = \frac{P - MC}{P} \times 100 \quad (8.10)$$

The optimal margin will be obtained by setting MC equal to MR. We can now obtain MR in terms of the PED, as follows:

$$\begin{aligned} MR &= \frac{dR}{dQ} = \frac{d(PQ)}{dQ} = P \frac{dQ}{dQ} + Q \frac{dP}{dQ} = P + Q \frac{dP}{dQ} = P + P \frac{Q}{P} \times \frac{dP}{dQ} \\ &= P \left(1 + \frac{Q}{P} \times \frac{dP}{dQ} \right) \\ MR &= P(1 + 1/\varepsilon) \end{aligned} \quad (8.11)$$

where ε represents the PED. Thus, assuming profit maximization, we can also write:

$$MC = P(1 + 1/\varepsilon) \quad (8.12)$$

We can now obtain the optimal profit margin in terms of the PED:

$$M = \frac{P - P(1 + 1/\varepsilon)}{P} = 1 - (1 + 1/\varepsilon) = -1/\varepsilon \quad (8.13)$$

This shows that products with more elastic demand should have a lower profit margin. Obviously, in perfect competition, when PED is infinite, there is no profit margin because $P = MC$.

Table 8.2. PED and mark-up

PED	Mark-up (%)
- 10	11
- 5	25
- 4	33
- 3	50
- 2	100
- 1.5	200
- 1	∞

b. Mark-up

Many students confuse margin with mark-up. Mark-up is defined as **the difference between the price and the marginal cost, expressed as a percentage of the marginal cost**. It can thus be written as:

$$U = \frac{P - MC}{MC} \times 100 \quad (8.14)$$

We have seen in (8.12) that at profit maximization $MC = P(1 + 1/\varepsilon)$. Therefore

$$P = \frac{MC}{1 + 1/\varepsilon} \quad (8.15)$$

We can also write:

$$P = MC(1 + U) \quad (8.16)$$

Therefore

$$\begin{aligned}
 MC(1 + U) &= \frac{MC}{1 + 1/\varepsilon} \\
 1 + U &= \frac{1}{1 + 1/\varepsilon}; \quad 1 + U = \frac{1}{\frac{\varepsilon+1}{\varepsilon}} = \frac{\varepsilon}{\varepsilon + 1} \\
 U &= \frac{\varepsilon}{\varepsilon + 1} - 1 = \frac{\varepsilon - (\varepsilon + 1)}{\varepsilon + 1} = \frac{-1}{\varepsilon + 1}
 \end{aligned} \quad (8.17)$$

Again we can see that products with more elastic demand should have a lower mark-up. The relationship between PED and mark-up is shown in Table 8.2.

It does not follow that firms or industries with higher margins and mark-ups are more profitable. For one thing, when managers refer to mark-ups they often do not use the same measure of cost that economists use. Some of these problems of cost measurement were discussed in Chapter 6. Managers, for example, often use some measure of average variable cost or average total cost, and do not take into account opportunity costs. This tends to result in mark-ups that are greater than true economic mark-ups. However, even

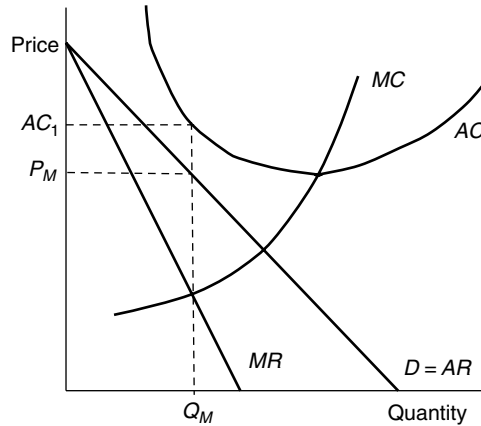


Figure 8.7. Loss-making monopoly.

when mark-up is measured in economic terms, a high mark-up does not necessarily indicate high profit, because it does not take into account the level of fixed costs. In some industries, fixed costs and mark-ups are very high. For example, in the airline industry capital costs are very high; in the breakfast cereal industry a very high proportion of revenue is spent on advertising and promotion (35 per cent); in the pharmaceutical industry huge amounts are spent on R&D. This leads us on to two common misconceptions regarding monopoly:

- 1 Monopolies always make large profits.
- 2 Monopolies have inelastic demand.

The first misconception can easily be seen to be incorrect by examining the performance of the state-run monopolies in the UK and elsewhere in Europe before privatization programmes began in the 1980s. These industries invariably made considerable losses, an issue examined in more detail in Chapter 12. A more theoretical approach is used in Figure 8.7. This graph is essentially the same as in Figure 8.6, but with the average cost curve shifted upwards. The equilibrium price and output are the same as before, but in this case a loss is made, given by $(AC_1 - P_M)Q_M$. This loss is unavoidable because the AC curve always lies above the demand curve; there is no output where the monopoly can cover its costs. Unless such a firm is state-subsidized it will not stay in business in the long run.

The second misconception, regarding monopolies having inelastic demand can be seen as false by recalling an important conclusion from Chapter 3: a firm will always maximize profit by charging a price where demand is elastic. We can also use Table 8.2 to see this; as demand approaches unit elasticity the optimal mark-up approaches infinity. There is therefore no reason for a firm to charge a price where demand has less than unit elasticity, in other words where demand is inelastic.

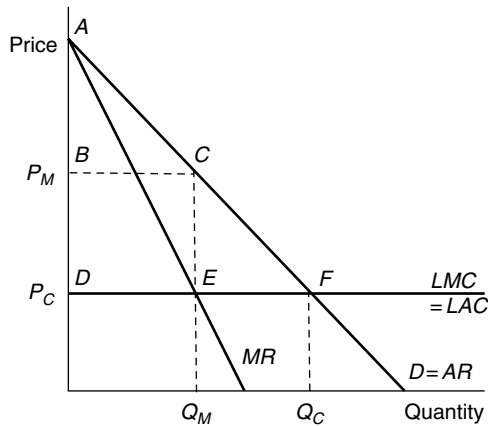


Figure 8.8. Comparison of perfect competition and monopoly.

8.3.6 Comparison of monopoly with perfect competition

There are four factors that can be compared here: price, output, profit and efficiency. It is helpful for analysis if both forms of market structure are shown on the same graph, as in Figure 8.8. This gives a long-run perspective. For the sake of simplicity it is assumed that long-run marginal costs are constant. This indicates that there are constant returns to scale, so that LMC and LAC are equal. P_M and Q_M represent the price and output of the monopolist and P_C and Q_C represent the price and total output of the industry in perfect competition (PC).

The factors listed above can now be examined in turn.

- Price.* In monopoly the price is higher than in PC.
- Output.* In monopoly the output is lower than in PC.
- Profit.* There is an element of supernormal profit in monopoly, given by the area of the rectangle $BCED$, although as we have just seen this is not always the case in monopoly. In perfect competition the price and long-run average cost are equal, resulting in only normal profit being made.

The fourth factor listed was *efficiency*. We now need to explain the difference between productive and allocative efficiency, the other type of efficiency mentioned earlier in the section on perfect competition.

1. *Productive efficiency.* In Figure 8.8 both the monopolist and the firm in PC are achieving productive efficiency, since they both have a constant level of LAC . However, if the monopolist has a rising LMC curve as in Figure 8.6, it will not be producing at the minimum point of its LAC curve, but at a point to the left of this. It will therefore not be achieving productive efficiency. The monopolist will be using too small a scale, and using it at less than optimal capacity.

2. *Allocative efficiency.* This refers to **the optimal allocation of resources in the economy as a whole**. In order to consider this aspect we need to introduce the concepts of consumer surplus and producer surplus. **Consumer surplus**

represents **the total amount of money that consumers are prepared to pay for a certain output over and above the amount that they have to pay for this output**. It is given by the area between the demand curve and the price line. Thus in perfect competition the consumer surplus is given by the area of triangle AFD in Figure 8.8.

Producer surplus, sometimes called economic rent, represents **the total amount of money that producers, meaning all factors of production, receive for selling a certain output over and above the amount that they need to receive to stay in their existing use in the long run**. It is given by the area between the marginal cost curve and the price line. In the special case shown in Figure 8.8 where MC is constant, producer surplus is equal to supernormal profit. However, if the MC curve is rising and there is perfect competition, the producer surplus will not be realized in the form of supernormal profit, since this will be competed away. The surplus will instead be distributed to the other factors of production, such as labour.

In order to examine and compare the allocative efficiency of the two types of market structure we need to consider the effects on total economic welfare of a change from perfect competition to monopoly. In perfect competition, total welfare is maximized because output is such that price equals marginal cost. This condition for allocative efficiency means that total welfare cannot be increased by any reallocation of resources; any gain for producers will be more than offset by a greater loss for consumers. In monopoly, output is such that price exceeds marginal cost, meaning that consumers would value any additional output more than it would cost the monopolist to produce it. However, it would not profit the monopolist to produce the additional output because their marginal revenue would fall below marginal cost. The total welfare loss can be seen in Figure 8.8. Although producers gain a surplus of BCED, as already mentioned, the size of the consumer surplus is reduced from AFD to ACB. This means that there is an **overall loss of welfare**, sometimes called a **deadweight loss**, of CFE.

One might ask at this point what relevance the total economic welfare aspects have for managerial decision-making; after all, managers are only concerned with the welfare of the firm. The reason for their relevance is that they affect government policy. As will be seen in more detail in Chapter 12, most governments monitor monopolistic industries and take an active role in discouraging restrictive practices. The impact of such policies on firms' strategies and profits can be considerable.

So far the picture painted of monopoly is an unfavourable one. However, to present a more balanced picture, it is necessary to stress that the analysis to this point has made some important and restrictive assumptions. For one thing, economies of scale have been ignored. In some industries, as seen in Chapter 6, these are of very great importance. Therefore, in industries like public utilities a monopoly may be able to produce more output more cheaply than firms in perfect competition, since firms can avoid the wasteful duplication of infrastructure like pipelines, railway tracks and cable lines.

Another factor ignored up to this point concerns the dynamic aspects of monopoly. Dynamic aspects relate to all the factors that influence economic change and growth over time. Most economists believe that such factors, such as R&D and innovation, are much more important than efficiency as far as long-run growth in productivity and living standards is concerned. In the comparative static analysis used above it is not possible to estimate the incentive effects that monopoly may have on R&D and innovation. Since a monopoly has the ability to profit from these over the long run, it may have a greater incentive to conduct R&D and develop new products than a firm in PC, which knows that any profit from such activities will rapidly be competed away. Empirical evidence regarding these aspects is somewhat inconclusive at present.

There now follows a case study on electricity generation, which explores in particular the relationships between cost structure and market structure, along with the impact of new technology.

Case study 8.1: Electricity generation

Here and now⁸

Distributed power generation will end the long-distance tyranny of the grid.

For decades, control over energy has been deemed too important to be left to the markets. Politicians and officials have been dazzled by the economies of scale promised by ever bigger power plants, constructed a long way from consumers. They have put up with the low efficiency of those plants, and the environmental harm they do, because they have accepted that the generation, transmission and distribution of power must be controlled by the government or another monopoly.

Yet in the beginning things were very different. When Thomas Edison set up his first heat-and-power co-generation plant near Wall Street more than 100 years ago, he thought the best way to meet customers' needs would be to set up networks of decentralised power plants in or near homes and offices. Now, after a century that saw power stations getting ever bigger, transmission grids spreading ever wider and central planners growing ever stronger, the wheel has come full circle. The bright new hope is micropower, a word coined by Seth Dunn of the WorldWatch Institute in an excellent report.* Energy prices are increasingly dictated by markets, not monopolies, and power is increasingly generated close to the end-user rather than at distant stations. Edison's dream is being revived.

The new power plants of choice the world over are using either natural gas or renewable energy, and are smaller, nimbler, cleaner and closer to the end-user than the giants of yesteryear. That means power no longer depends on the vagaries of the grid, and is more responsive to the needs of the consumer. This is a compelling advantage in rich countries, where the digital revolution is fuelling the thirst for high-quality, reliable power that the antiquated grid seems unable to deliver. California provides the best evidence: although the utilities have not built a single power plant over the past decade, individuals and companies have added a whopping 6gW of non-utility micropower over that period, roughly the equivalent of the state's installed nuclear capacity. The argument in favour of micropower is even more persuasive in developing countries, where the grid has largely failed the poor.

This is not to say that the existing dinosaurs of power generation are about to disappear. Because the existing capital stock is often already paid for, the marginal cost of running existing power plants can be very low. That is why America's coal-fired plants, which produce over half the country's power today, will go on until the end of their useful lives, perhaps decades from now – unless governments withdraw the concessions allowing them to exceed current emissions standards.

While nobody is rushing to build new nuclear plants, old ones may have quite a lot of life left in

them if they are properly run, as the success of the Three Mile Island nuclear power plant in Pennsylvania attests. After the near-catastrophic accident in 1979 that destroyed one of the plant's two reactors, the remaining one now boasts an impressive safety and financial record. Safety and financial success are intimately linked, says Corbin McNeill, chairman of Exelon and the current owner of the revived plant. He professes to be an environmentalist, and accepts that nuclear power is unlikely to be the energy of choice in the longer term: 'A hundred years from now, I have no doubt that we will get our energy using hydrogen.' But he sees nuclear energy as an essential bridge to that future, far greener than fossil fuels because it emits no carbon dioxide.

GOOD OLD GRID

The rise of micropower does not mean that grid power is dead. On the contrary, argues CERA, a robust grid may be an important part of a micropower future. In poor countries, the grid is often so shoddy and inadequate that distributed energy could well supplant it; that would make it a truly disruptive technology. However, in rich countries, where nearly everyone has access to power, micropower is much more likely to grow alongside the grid. Not only can the owners of distributed generators tap into the grid for back-up power, but utilities can install micropower plants close to consumers to avoid grid bottlenecks.

However, a lot of work needs to be done before any of this can happen. Walt Patterson of the Royal Institute of International Affairs, a British think-tank, was one of the first to spot the trend toward micropower. He argues that advances in software and electronics hold the key to micropower, as they offer new and more flexible ways to link parts of electricity systems together. First, today's antiquated grid, designed when power flowed from big plants to distant consumers, must be upgraded to handle tomorrow's complex, multi-directional flows. Yet in many deregulated markets, including America's, grid operators have not been given adequate financial incentives to make these investments. To work effectively, micropower also needs modern command and communications software.

Another precondition is the spread of real-time electricity meters to all consumers. Consumers

who prefer stable prices will be able to choose hedged contracts; others can buy and sell power, much as day traders bet on shares today. More likely, their smart micropower plants, in cahoots with hundreds of others, will automatically do it for them.

In the end, though, it will not be the technology that determines the success of distributed generation, but a change in the way that people think about electricity. CERA concludes that for distributed energy, that will mean the transition from an equipment business to a service business. Already, companies that used to do nothing but sell equipment are considering rental and leasing to make life easier for the user.

Forward-looking firms such as ABB, a Swiss-Swedish equipment supplier, are now making the shift from building centralised power plants to nurturing micropower. ABB is already working on developing 'microgrids' that can electronically link together dozens of micropower units, be they fuel cells or wind turbines.

Kurt Yeager of the Electric Power Research Institute speaks for many in the business when he sums up the prospects: 'Today's technological revolution in power is the most dramatic we have seen since Edison's day, given the spread of distributed generation, transportation using electric drives, and the convergence of electricity with gas and even telecoms. Ultimately, this century will be truly the century of electricity, with the microchip as the ultimate customer.'

* 'Micropower: the next electrical era', by Seth Dunn. WorldWatch Institute, 2000.

Questions

- 1 Explain why power generation has traditionally been a monopoly in all developed countries.
- 2 What is meant by a transmission grid? How is this feature related to a monopolistic market structure?
- 3 What is meant by micropower? What are its implications for grid systems?
- 4 What are the implications of micropower for the environment?
- 5 How do you think changes in technology will affect the market structure of the power generation industry?

8.4 Monopolistic competition

Although economics textbooks tend to concentrate more on discussing perfect competition and monopoly, monopolistic competition and oligopoly are more prevalent in practice. The theory of monopolistic competition, as an intermediate form of market structure between perfect competition and monopoly, was originally developed by Chamberlin⁹ in 1933. Its characteristics were summarized in Table 8.1 and we can now examine the conditions for monopolistic competition in more detail.

8.4.1 Conditions

There are five main conditions for monopolistic competition to exist:

- 1 There are many buyers and sellers in the industry.
- 2 Each firm produces a slightly differentiated product.
- 3 There are minimal barriers to entry or exit.
- 4 All firms have identical cost and demand functions.
- 5 Firms do not take into account competitors' behaviour in determining price and output.

As far as the first condition is concerned, there may be a few large dominant firms with a large fringe of smaller firms, or there may be no very large firms but just a large number of small firms. Grocery retailing is an example of the first situation, while the car repair industry is an example of the second. In both cases there is product differentiation, and the significance of this is that firms are not price-takers, but, rather, have some control over market price. However, this control is not as great as that of the monopolist for two reasons. First the firms' products have closer substitutes than the product of a monopolist, making demand more elastic. The second reason is related to the third condition above: the low barriers to entry mean that any supernormal profit is competed away in the long run. This also involves the fourth condition, that firms have identical cost curves. We can now examine this situation graphically.

8.4.2 Graphical analysis of equilibrium

In the short run the equilibrium of the firm in monopolistic competition is very similar to that of the monopolist. Profit is again maximized by producing the output where $MC = MR$. Supernormal profit can be made, depending on the position of the AC curve, because the number of firms in the industry is fixed. The only real difference between the two situations is that in Figure 8.9, relating to monopolistic competition, the demand curve (and hence the MR curve) is flatter than the demand curve in Figure 8.6 relating to monopoly. This is because of the greater availability of substitutes.

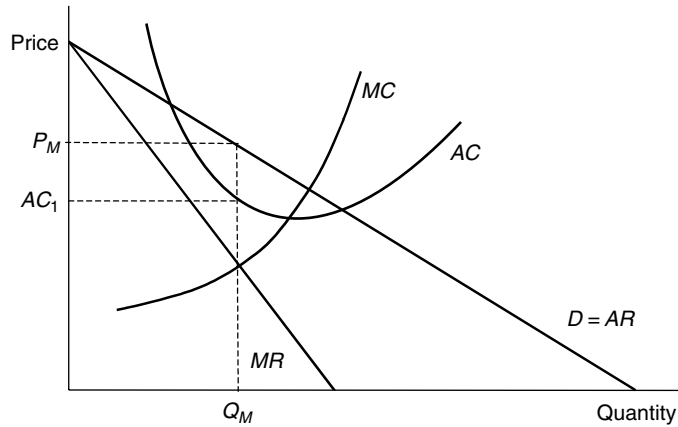


Figure 8.9. Short-run equilibrium for firms in monopolistic competition.

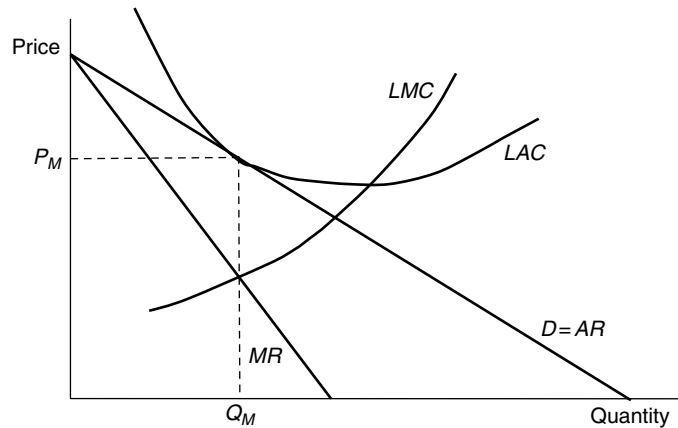


Figure 8.10. Long-run equilibrium for firms in monopolistic competition.

In the long run, new firms will enter the industry, attracted by the supernormal profit. This will have the effect of shifting the demand curve downwards for existing firms. The downward shift will continue until the demand curve becomes tangential to the AC curve (LAC in this case), at which point all supernormal profit will have been competed away. This situation is illustrated in Figure 8.10.

8.4.3 Algebraic analysis of equilibrium

Let us take a market where firms have the following demand and cost functions:

$$\text{Demand: } P = 140 - 4Q$$

$$\text{Cost: } TC = 120Q - 12Q^2 + 2Q^3$$

where P is in £, TC is in £ and Q is in thousand units.

In this case the total cost function is a cubic function, with no fixed costs (for the sake of simplicity).

$$MR = 140 - 8Q$$

$$MC = 120 - 24Q + 6Q^2$$

$$140 - 8Q = 120 - 24Q + 6Q^2$$

$$6Q^2 - 16Q - 20 = 0$$

$$Q = \frac{16 \pm \sqrt{16^2 - (4)(6)(-20)}}{12} = 3.594 = \mathbf{3,594 \text{ units}}$$

$$P = \mathbf{£125.60}$$

$$\text{Profit} = (P - AC)Q$$

$$AC = 120 - 12(3.594) - 2(3.594)^2 = £102.70$$

$$\text{Profit} = (125.6 - 102.7)3.594 = 82.30 = \mathbf{£82,300}$$

This profit will attract new entrants into the industry in the long run, causing the demand curve for existing firms to shift downwards. We will assume that this is a parallel shift, with no change in slope; in other words, there is no change in the marginal effect of price on sales. We will also assume for simplicity that the long-run cost curves are identical to the short-run curves. The new demand curve faced by firms is given by:

$$P = a - 4Q \text{ where } a \text{ is a constant to be determined}$$

$$MR = a - 8Q$$

$$MC = 120 - 24Q + 6Q^2 = a - 8Q$$

$$a = 120 - 16Q + 6Q^2$$

In long-run equilibrium, $P = AC$, as all supernormal profit is competed away.

$$a - 4Q = 120 - 12Q + 2Q^2$$

$$a = 120 - 8Q + 2Q^2$$

Thus:

$$120 - 16Q + 6Q^2 = 120 - 8Q + 2Q^2$$

$$4Q^2 = 8Q$$

$$Q = 2 = \mathbf{2,000 \text{ units}}$$

$$a = 120 - 16(2) + 6(2)^2 = 112$$

New demand curve is:

$$P = 112 - 4Q$$

$$P = 112 - 4(2) = \text{£}104$$

8.4.4 Comparison with perfect competition and monopoly

There are four areas where comparison can be made:

- a. *Price*. This tends to be higher than in perfect competition (PC), being above the minimum level of average cost, in both the short run and the long run (similar to monopoly).
- b. *Output*. This tends to be lower than in PC, since firms are using a less than optimal scale, at less than optimal capacity (similar to monopoly).
- c. *Productive efficiency*. This is lower than in PC, for the reason stated previously.
- d. *Allocative efficiency*. There is still a net welfare loss, because $P > MC$.

It is to be noted that even though no supernormal profit is made in the long run, neither productive nor allocative efficiency is achieved. This has led a number of people to criticize the marketing function of firms. This activity creates product differentiation and is thus claimed to cause inefficiency. In order to evaluate this argument one would also have to assess the benefits of the marketing function in terms of increasing customer awareness and knowledge, and reducing transaction costs.

Some of the assumptions involved in the above analysis should be examined at this stage. The last three conditions in subsection 8.4.1 are all questionable in terms of their realism. Entry and exit barriers may be low rather than non-existent, and some firms may be more efficient than others. Thus the relaxation of the third and fourth assumptions may result in some firms being able to make supernormal profit in the long run. Only the marginal firm, meaning the least efficient firm, may in fact just be making normal profit, while all other firms make some amount of supernormal profit, depending on their efficiency and the level of entry and exit barriers.

8.4.5 Comparison with oligopoly

It is the last assumption, regarding the independence of firms' decision-making that has attracted the most attention from economists. Many economists claim for example that monopolistic competition is not really a distinct form of market structure.^{10,11} This claim is based on the observation that firms are typically faced with competition from a limited number of neighbouring firms, with markets being segmented spatially. Segmentation may also be in terms of product characteristics.

An example will illustrate this situation. The restaurant industry is not a single market. An individual restaurant does not compete with all other

restaurants in the country, or even in the same town. It may compete with other restaurants within a one-mile radius; furthermore, it may not compete strongly with some of these restaurants because they are not seen as being close substitutes. Thus an Indian restaurant may not compete with Italian, French, Greek or Mexican restaurants to any great degree. This degree of competition can be examined empirically by measuring the cross elasticity of demand. Of course, it can be argued that if there are few competitors then the product is not *slightly* differentiated, as required in monopolistic competition, but highly differentiated. However, regardless of how the assumptions of monopolistic competition are violated, it seems that in view of these factors it is often preferable to consider firms in many situations as being involved in a system of intersecting oligopolies, with low entry and exit barriers.

It is to oligopoly that we must now turn our attention. However, before doing this, it is useful to consider a situation that has received much attention in the media lately, since it relates to price-fixing and cartels. The situation is examined in more detail in Chapter 12, when competition policy is discussed, but it is appropriate to consider certain aspects in the current context.

Case study 8.2: Price cuts for medicines

Chemists at risk as prices are slashed¹²

BY NIGEL HAWKES, HEALTH EDITOR

Big price cuts on a wide range of medicines and vitamins were promised by the supermarket chains yesterday as 30 years of price-fixing were swept aside.

Many popular products, including painkillers, cough medicines, indigestion tablets and nutritional supplements are being halved in price from last night, with reductions of between 20 and 40 per cent on many others.

The Office of Fair Trading called it excellent news for consumers but the body representing small pharmacies said that many would close, threatening community services.

The big supermarkets trumpeted 'millions of pounds-worth of savings' as they competed to offer the biggest reductions. At Asda, a packet of 16 regular Anadin will be 87p, instead of £1.75, and Nurofen tablets will cost £1.14 for 16, rather than £2.29.

Reductions at Tesco included a 40 per cent cut in Anadin Extra, to £1.29 for 16, while Sainsbury's matched the Asda price for Nurofen, and reduced Seven Seas Evening Primrose Oil from £5.59 for a 60-pack to £2.79.

The cuts came after the Community Pharmacy Action Group, representing small retailers, withdrew

its opposition to a High Court action brought by the Office of Fair Trading. The OFT had sought the abolition of resale price maintenance in the industry, exempted 30 years ago from general price-fixing rules to try to ensure the survival of small pharmacies. There are 13,500 pharmacies in Britain, of which 9,000 are small shops serving local high streets and rural communities.

The action group backed out after Mr Justice Buckley said that he believed there was insufficient proof that a large number of independent pharmacies would close, or that the range of products would be reduced. But the group's chairman, David Sharpe, said that the outcome would be a devastating blow. 'Many pharmacists will simply not be able to survive given the buying power and aggressive pricing of the supermarkets' he said. 'It's a sad day for Britain. The potential losers are the elderly, disabled and young mothers who rely on the free advice and range of services offered by the local pharmacist. We'll fight on and hope the public will remain loyal.'

The changes will cover about 2,500 products sold without requiring a doctor's prescription, and will have no effect on prescription drugs or on cosmetics sold by pharmacists.

Prices are likely to fall even lower as competition grows. In the United States, where prices are

unregulated, comparable products are markedly cheaper.

Richard Hyman, chairman of the Verdict retail research consultancy, said: 'This is a market made for supermarkets. Medicines are small, they fit on shelves and supermarkets are going to make a lot of noise about the great prices that they will be offering. Soon medicines will become like any other product and be part of the weekly shop.'

John Vickers, Director-General of Fair Trading, said: 'This is excellent news for consumers, who will now benefit from lower and more competitive prices for common household medicines. Consumers will save many millions of pounds a year.'

The Proprietary Association of Great Britain, which represents medicine and food supplement manufacturers, said it was disappointed.

Questions

- 1 What kind of market structure is involved for the sale of medicines and vitamins?
- 2 What can be said about barriers to entry in this market?
- 3 Might there be a change in market structure after the change in the law?
- 4 Explain the disadvantages of the abolition of resale price maintenance (RPM) for this market.
- 5 When RPM was abolished for book sales in 1995, the same concerns as those expressed in the above case were voiced. Since then, 10 per cent of bookshops have gone out of business. What conclusions might this help you to draw regarding the future of small pharmacies?
- 6 How does the rise of the Internet affect this situation?

8.5 Oligopoly

An oligopolistic market structure describes the situation where a few firms dominate the industry. The product may be standardized or differentiated; examples of the first type are steel, chemicals and paper, while examples of the second type are cars, electronics products and breakfast cereals. The most important feature of such markets that distinguishes them from all other types of market structure is that firms are interdependent. Strategic decisions made by one firm affect other firms, who react to them in ways that affect the original firm. Thus firms have to consider these reactions in determining their own strategies. Such markets are extremely common for both consumer and industrial products, both in individual countries and on a global basis. However, there is a considerable amount of heterogeneity within such markets. Some feature one dominant firm, like Intel in computer chips; some feature two dominant firms, like Coca-Cola and Pepsi in soft drinks; some feature half a dozen or so major firms, like airlines, mobile phones or athletic footwear; and others feature a dozen or more firms with no really dominant firm, like car manufacturers, petroleum retailers, and investment banks. Of course, in each case the number of major firms depends on how the market is defined, spatially and in terms of product characteristics.

8.5.1 Conditions

The main conditions for oligopoly to exist are therefore as follows:

- 1 A relatively small number of firms account for the majority of the market.
- 2 There are significant barriers to entry and exit.
- 3 There is an interdependence in decision-making.

As far as the first condition is concerned there are a number of measures that are used to indicate the degree of **market concentration** in an industry. The easiest to interpret are the four-firm or eight-firm **concentration ratios**. These indicate **the proportion of the total market sales accounted for by the largest four or eight firms in the industry**. A more detailed measure, though more difficult to interpret, is the **Herfindahl index**. This index is computed by taking **the sum of the squares of the market shares of all the firms in the industry**. For example, if two firms account for the whole market on a 50:50 basis, the Herfindahl index (H) would be $(0.5)^2 + (0.5)^2 = 0.5$. In general terms, the index is given by:

$$H = \sum S^2 \quad (8.18)$$

where S = the proportion of the total market sales accounted for by each firm in the industry. A value of this index above 0.2 normally indicates that the market structure is oligopolistic. Another measure, related to the Herfindahl index but easier to interpret, is the **numbers-equivalent of firms** (NEF); this is given by **the reciprocal of the H value**. It corresponds to the number of firms that would exist in the industry, given a certain value of H , assuming that all firms had an equal market share. We can see in the example of two firms above that the $NEF = 1/0.5 = 2$. Thus an industry with an H -value of 0.2 corresponds to a situation where the market is shared by five firms equally.

In order for oligopolies to evolve and maintain their market structure there must be significant barriers to entry and exit. These barriers have been discussed in the section on monopoly, but economies of scale, sunk costs and brand recognition are all important. Such barriers prevent or discourage the entry of new firms and allow existing firms to make supernormal profit, even in the long run. The strategic use of these barriers is discussed in the next chapter.

The third condition above, interdependence in decision-making, is also discussed in the next chapter. This is because the kind of analysis involved is different from that so far discussed, and is heavily dependent on a branch of decision theory known as game theory. This requires a fairly lengthy exposition in order to convey the important principles, and some of the material is advanced in nature. Therefore, it is important to stress that this chapter will present only a brief overview of the essential aspects of oligopolistic market structures. This means that the only model introduced at this stage is a highly simplified and in many ways incomplete model. The more complex and realistic models are developed in the next chapter.

8.5.2 The kinked demand curve model

This model was originally developed by Sweezy¹³ and has been commonly used to explain **price rigidities** in oligopolistic markets. A price rigidity refers to a situation where **firms tend to maintain their prices at the same level in spite**

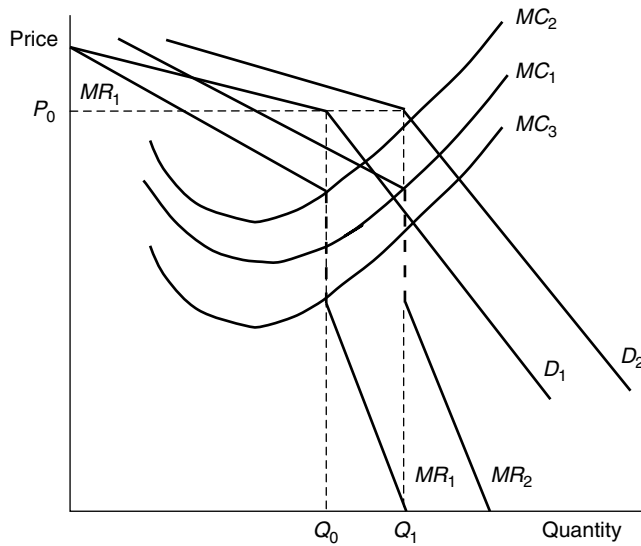


Figure 8.11. The kinked demand curve and price rigidity.

of changes in demand or cost conditions. The model assumes that if an oligopolist cuts its prices, competitors will quickly react to this by cutting their own prices in order to prevent losing market share. On the other hand, if one firm raises its price, it is assumed that competitors do not match the price rise, in order to gain market share at the expense of the first firm. In this case the demand curve facing a firm would be much more elastic for price increases than for price reductions. This results in the kinked demand curve shown in Figure 8.11. It should be noted that this is not a 'true' demand curve as defined in Chapter 3, since it no longer assumes that other things remain equal, apart from the price charged by the firm. If the price charged falls below P_0 , it is assumed that other firms react to this and reduce their own prices. We might call it an 'effective' demand curve.

The kink in the demand curve causes a discontinuity or break in the MR curve. The consequence of this is that if the marginal cost function shifts from the original function MC_1 upwards or downwards within the range from MC_2 to MC_3 , then the profit-maximizing output will remain at Q_0 and the price will remain at P_0 , since the MC curve passes through the MR curve in the vertical break. Similarly, if the demand curve shifts from D_1 to D_2 , the MR curve will shift to the right to MR_2 , but the original MC curve will still pass through the vertical break. This means that the profit-maximizing output will increase from Q_0 to Q_1 , but the price will remain the same at P_0 . The reason for this is that the vertical break occurs below the kink in the demand curve, which is at the prevailing price P_0 .

The above model can be criticized on three main grounds:

- 1 It takes the prevailing price as given; there is no attempt to explain how this prevailing price is determined in the first place.

- 2 It makes unrealistic assumptions regarding firms' behaviour in terms of following price increases. It will be seen in the next chapter that there may be good reasons for following a price increase as well as following a decrease.
- 3 Empirical evidence does not generally support the model.¹⁴ As mentioned above, in reality firms tend to follow price increases just as much as they follow price reductions.

In spite of the above shortcomings the kinked demand curve model remains a popular approach to analysing oligopolistic behaviour. For one thing, it suggests that firms are likely to co-operate on the monopoly price, and this fact is easily observed in practice. We now need to turn our attention to such co-operation.

8.5.3 Collusion and cartels

Collusion is the term frequently used to refer to **co-operative behaviour** between firms in an oligopolistic market. Such collusion may be explicit or tacit; this chapter discusses the first type of collusion, whereas the next chapter discusses the second type. Explicit collusion often involves the firms forming a **cartel**. This is **an agreement among firms, of a formal or informal nature, to determine prices, total industry output, market shares or the distribution of profits**. Most such agreements are illegal in developed countries, though they are still widely practised on an informal basis because their existence is difficult to prove. In some cases, cartels are actually encouraged and protected by governments, for example the various agricultural marketing boards in the UK and in other countries. There are also producers' associations, for example representing taxi drivers, which may have the legal right to restrict entry into the industry, at least on a local scale. On an international basis the best-known cartel is OPEC, the Organization of Petroleum Exporting Countries, which has existed for decades with a mixed record of success for its members. The most important issues to discuss regarding cartels are first the incentives to form them, and second the factors determining their likely success.

a. Incentives

Firms in an oligopolistic market structure can increase profit by forming a cartel. This is most easily explained by considering a simple example. Let us take an industry producing a standardized product, with just two firms; the market demand curve is $P = 400 - 2Q$, with each firm having a constant marginal cost of £40 and no fixed costs. This situation is shown in Figure 8.12. Essentially the situation is similar to that in Figure 8.8, comparing perfect competition and monopoly. If the two firms compete in price (so-called Bertrand competition), the price will be forced down to the level of marginal cost. This is because each firm can grab 100 per cent of the market share by undercutting the competitor, so this undercutting will continue until all

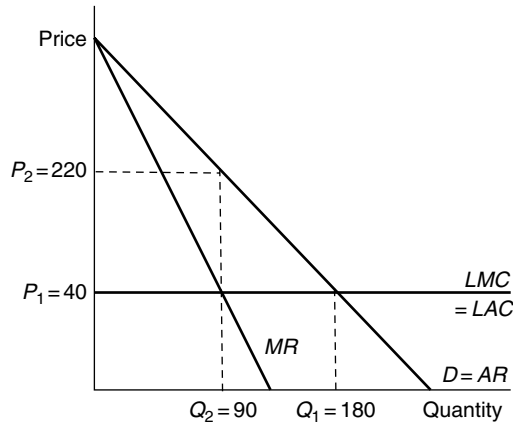


Figure 8.12. Effects of a cartel.

supernormal profit is competed away. Obviously the price will be £40 in this case, and the total market output will be 180 units (from the demand equation).

If the firms form a cartel they can charge the monopoly price. In order to determine this we have to determine the output where $MC = MR$. This is done as follows:

$$P = 400 - 2Q$$

$$R = 400 - 2Q^2$$

$$MR = 400 - 4Q$$

$$MC = 40$$

$$400 - 4Q = 40$$

$$4Q = 360, Q = 90$$

$$P = 400 - 2(90) = \textbf{£220}$$

In this case the industry will make a profit given by

$$(P - AC)Q = (220 - 40)90 = \textbf{£16,200}.$$

Thus, assuming that the profits are shared equally, each firm can make a profit of £8,100. This is clearly preferable to the competitive situation. At this stage we are ignoring the more complicated situation where the firms compete in terms of output by considering what output the other firm will put on the market. This is called Cournot competition, and both Cournot and Bertrand competition are discussed in the next chapter.

Although both firms can make supernormal profit by forming a cartel, this profit can only be sustained if the firms agree to restrict total output. This usually involves setting output quotas for each firm; in the above example the quotas would be 45 units each. The enforcement of output quotas creates a problem for cartels; each member firm can usually profit at the expense of the

others by 'cheating' and producing more than its output quota, thus making the cartel unstable. We now need to consider the factors that affect the likelihood of success of a cartel.

b. Factors affecting success of a cartel

There are a number of factors that are relevant, and the most important ones are examined here.

1. Number of sellers. As the number of sellers increases it is more likely that individual firms will ignore the effects of their pricing and output on other firms, since these will be smaller. A big increase in one firm's output will not have as much effect on the industry price when there are a dozen firms in the industry than when there are just two firms. Furthermore, firms are more likely to have disagreements regarding price and output strategies if there are more firms, and therefore they are again more likely to act independently. This has been a problem for OPEC because of the relatively large number of members. OPEC's problems are increased because, having once controlled 55 per cent of world oil output, it currently only controls less than 30 per cent.

2. Product differentiation. Co-operation is easier for firms if they are producing a homogeneous or standardized product, because in this case the firms can only compete in terms of price. With differentiated products competition can occur over a whole array of product characteristics. Even with a product like crude oil there is not complete homogeneity; there are different grades according to country of origin, and sellers can also vary payment terms as a form of competition.

3. Cost structures. As with differences in product, differences in cost structures can make co-operation more difficult. Co-operation is also more difficult in capital-intensive industries where fixed costs are a high proportion of total costs. This is because if firms are operating at less than full capacity it is possible to increase profits considerably by increasing output and cutting prices.

4. Transparency. If the market is transparent it will not be possible for a firm to undercut its competitors secretly. Cartels may therefore take steps to publicize information regarding the transactions of members in order to prevent them from conducting secret negotiations. However, it may still be possible to hide certain details of transactions, such as payment terms, which in effect can amount to a price reduction.

Because many of the above characteristics have not been favourable, many cartels have proved to be unstable in practice, and have been short-lived. Again, the stability of cartels is examined in the next chapter, since the behaviour of the members tends to conform to a repeated game. Some cartels in Europe that have in the past enjoyed government protection, for example the coal and steel industries, are now also in trouble; recent pressures related to

competition in the so-called single market have undone much of this valued protection. However, recent protectionist measures by the US administration involving these industries and others may change this picture to some extent. These aspects of government policy will be examined in more detail in Chapter 12.

8.5.4 Price leadership

A commonly observed pattern of behaviour in oligopolistic industries is the situation where one firm sets a price or initiates price changes, and other firms follow the leader with a short time lag, usually just a few days. There are various ways in which such behaviour can occur, depending on two main factors.

1. *Product differentiation.* For homogeneous products the followers normally adjust their prices to the same level as the leader. In the more common case of differentiated products the price followers generally conform to some structure of recognized price differentials in relation to the leader. Thus Ford may adjust the prices of various models so that they are given percentages lower or higher than some benchmark GM model.

2. *Type of leadership.* There are two main possibilities here. **Dominant** price leadership refers to the situation where the price leader is usually the largest firm in the industry. In this case the leader is fairly certain of how other firms will react to its price changes, in terms of their conforming to some general price structure, as described above. This certainty may be increased by the implicit threat of retaliation if a competitor does not follow the leader. The other main type of price leadership is called **barometric**. This time the price leader is not necessarily the largest firm, and leaders may frequently change. There is more uncertainty in this case regarding competitive reactions, but the leader is normally reacting to changes in market demand or cost conditions, and suggesting to other firms that it is in their interests to follow the changes.

When prices and outputs are determined under a price leadership situation there is no explicit collusion among firms, even though the almost simultaneous price changes may cause consumers and regulatory authorities to be suspicious. In the next chapter we shall see what makes firms conform to such changes so quickly. The following case studies examine the factors which can affect the success of collusion in two industries that have been exposed to the media spotlight.

Case study 8.3: Mobile phone networks

Predatory roaming¹⁵

They were in the bank, toting guns, as lots of money happened to go from the vault. That was the essence of last week's claim by Mario Monti, the European Union's competition commissioner, that mobile-

phone operators have gouged customers by colluding to raise rates for roaming – ie, when you use your mobile phone abroad. Mr Monti's case is circumstantial, but he says the network operators will have to answer it.

In December Mr Monti's office issued a report on the market for roaming. Most countries in the European Economic Area (EEA), the report found, have a roaming market that is ripe for collusion. The product is undifferentiated, and the number of sellers small. Pricing in the wholesale market is transparent, making it easy for a market leader to raise prices, and for other operators to take the hint and follow suit. The costs of running mobile networks do not vary that much. As a result, says the report, sellers' pricing structures tend to run in parallel, at 'high and rigid' levels. Mr Monti cites 'an almost complete absence of competition', and says that 'prices appear to be converging', towards €1 (89 cents) a minute.

To be fair, the conditions for collusion, apart from the small number of sellers cited above, could also be present in a perfectly competitive market. And retail prices in Europe are not quite as similar as Mr Monti's comments suggest. For a call from Belgium to Britain today, using a British mobile phone, rates range from 51p (73 cents) to 99p a minute. Rates for receiving calls also vary widely. On One2One, a monthly charge of only £2.50 can lower the receiving rate from 76p to 16p. That is an indication of just how low the marginal cost of roaming calls might be.

Looking closely at wholesale rates, the commission found that the cheapest in Europe were about €0.46 a minute. In Belgium, Britain, the Netherlands and Norway, some operators had rates at least twice as high as the average of the five cheapest. Yet even the lowest wholesale rates in Europe may be gouging consumers. Just look at what is on offer in North America. MicroCellnet, a Canadian operator that has 1m customers, recently launched a flat-rate American roaming service: for customers on a standard monthly service agreement, the retail price of calls made anywhere to Canada or within the United States is 20 cents a minute – less than half even the lowest wholesale rates in Europe.

Perhaps Europe's costs are so different from North America's that they justify BT Cellnet's roaming rate of

99p a minute? It seems unlikely. Chris Doyle, an economist at Charles River Associates, points out that roaming generates up to 35% of European operators' revenues, although it accounts for a much smaller share of the time customers spend on the telephone. Asked exactly what costs and market forces determine its roaming rates, BT Cellnet says the question is 'too commercially sensitive to answer'.

Market concentration also points to a lack of competition. In each of 11 EEA countries, a single operator had a market share of at least 50%. Still, the biggest obstacle to a competitive market for roaming may be the ease with which the operators can exploit consumers. They have little incentive to compete over roaming rates – to quit the cartel, Mr Monti might say – since mobile users do not usually use rates abroad as a basis for choosing a provider. Few customers know how much they are paying for roaming. Even fewer actively choose which local network to roam on.

The commission's report recommends making choice easier for consumers. In the best of worlds, roamers would be able to get rate information piped through to their telephones from various providers, before choosing which service to use. Mr Doyle believes that call-back services, which allow roamers to replace higher calling fees with lower receiving fees, will put pressure on operators to cut rates. If the commission wants to see rates fall swiftly, however, it will have to take action itself.

Questions

- 1 Why is the roaming market in the EEA 'ripe for collusion'?
- 2 What is the nature of the barriers to entry in the market?
- 3 Why is it easy for the operators to exploit consumers in this case?
- 4 If the commission does not take action, do you think it is likely that rates will fall much in the future?

Case study 8.4: Private school fees

Private schools in row over fee-fixing¹⁶

Some of Britain's top private schools stand accused of price-fixing after meeting to plan steep increases in fees, which lawyers say could breach competition laws.

Eton, Westminster and Marlborough are among the schools that appear to have colluded on the fees they charge. The Office of Fair Trading is now considering launching an investigation as parents

face record hikes in fees averaging 10%, four times the rate of inflation. In the past decade fees have risen by 56%.

Across the country, local and national groups of schools have 'cartel-style' private meetings where they share sensitive financial information. The result is near-identical increases in fees.

One bursar admitted last week that he had shared pricing information with other schools and compiled a dossier of his rivals' future fees that would be presented to his governing body before finalising his own.

David Chaundler, bursar at Westminster school, said he acquired details of rivals' fee proposals and costs from meetings of the Eton Group of 12 top private schools. At one meeting in February each bursar announced their school's proposals for increasing fees.

'We do compare school fees,' Chaundler said. 'If I went to my governors with a rise substantially above the others they might tell me to rethink. We do ensure we are pretty well in line.'

Competition lawyers believe the relationship could constitute a cartel. Jonathan Tatten, a partner at Denton Wilde Sapte, said the schools, which have charitable status and are non-profit-making, were not exempt from competition laws: 'Showing confidential pricing information to competitors is a very serious breach of competition rules. You know where you can safely pitch your own fees and it's a way of fixing the market.'

The maximum punishment if a cartel is found is a five-year prison term, he added. In America, a price-fixing inquiry into Ivy League colleges ended without any principals going to jail but led to new rules banning discussion of fees with each other.

Westminster and the other schools say they still make independent decisions on the precise level of fees, and claim the prices are close because many schools have similar cost bases.

This year private schools face a financial crunch from higher salaries and pension payments for teachers, plus Gordon Brown's rise in National Insurance contributions. Fearing a backlash from parents against big fee rises, this spring schools were particularly keen to present a united front. Top boarding schools are set to cross the £20,000-a-year fees watershed for the first time.

On February 7 the Eton Group, including Westminster, Marlborough, King's College school (London), Sherborne, Tonbridge and Bryanston, met at Dulwich College, south London. Each of the bursars outlined the fees they proposed to charge for the next year.

Andrew Wynn, of Eton, admitted: 'We do meet and talk about fees to get some idea of what other schools are thinking. We are a co-operative bunch, and we are not out to slit each other's throats.'

Although their academic results vary, the group's six provincial boarding schools are already closely aligned on fees of £6,300 to £6,445 a term. Its two major London boarding schools, Westminster and Dulwich, charge fees of more than £6,000 and are just £138 apart. Day school members Highgate and nearby University College school have charged exactly the same for the past two years.

A similar meeting held by a rival network, the Rugby Group, whose members include Winchester, Radley, Harrow, Clifton College and Shrewsbury, is also understood to have discussed plans for the first £20,000 annual fees.

William Organ, bursar of Winchester, said: 'Sometimes schools feel they are too far ahead in fees and row back a bit, or the other way round. They look at their competitors in the area and say: Gosh, we're slipping behind in the fees league we'd better catch up.'

A network of six leading private day schools in Manchester, known as the Consortium, holds similar meetings. The schools including Manchester Grammar, William Hulme's Grammar and Stockport Grammar, last year charged about £1,900 a term, with a difference of £131 between them.

Elizabeth Fritchley, William Hulme's bursar, said the group met every term and phoned each other in March: 'We decide what our increase is to be and then phone the other schools. If we are thinking of putting the fees up by, say, 15% and the rest were proposing far less, then it would make us rethink our strategy.'

Yesterday Mike Sant, general secretary of the Independent Schools' Bursars Association, denied any cartels were operating: 'Schools will decide where they want to be in the market and will be watching their competition and move fees accordingly. All the schools are so different they are just not in

competition. They do exchange information, but just to get a feel for what others are doing.'

Questions

- 1 If a group of schools simultaneously raises their fees by a similar amount, is this evidence of collusion? What other explanation might be possible?
- 2 If fees have risen much faster than the rate of inflation, 56 per cent over a decade, is this evidence of collusion, or are other explanations possible?
- 3 Describe the factors that are favourable to the formation of a successful cartel, and those that are unfavourable, for the elite private schools mentioned in the article. Use the statements in quotations as evidence.

8.6 A problem-solving approach

The essential problem in the issue of market structure is the determination of price and output, given the different market conditions involved. Conclusions relating to profit and efficiency follow from this. The starting point is always the **demand and cost functions**. In some situations that the student may face these will not be given in equation form, as for example in Problem 8.1. The first step in that case is to derive the demand and cost functions from the information given. Once this is done there is a straightforward **five-step procedure** to solving the problem. There are no additional solved problems in this chapter because examples of these have already been given in the text, under each form of market structure. The student should be able to see from these examples that in each case the following general steps are involved:

- 1 Derive demand function in the form $P=f(Q)$.
- 2 Derive revenue function in the form $R=f(Q)$.
- 3 Derive marginal revenue function in the form $MR=\frac{dR}{dQ}=f(Q)$.
- 4 Derive marginal cost function in the form $MC=\frac{dC}{dQ}=f(Q)$.
- 5 Set $MC=MR$ and solve for Q .

Once the value of Q is obtained the value of P can be obtained from the demand equation. Profit can be calculated either by taking revenue minus costs, or by using the equation:

$$\text{Profit} = (P - AC)Q$$

The above procedure is very robust and can be used with any mathematical form of demand and cost function. The algebra may vary, as seen for example in the case of monopolistic competition, where a cubic cost function is used, but in each problem the general procedure is identical.

It should be noted at this stage that the above procedure is not the only approach that can be used for solving problems. Another approach is to **derive the profit function** for the firm, in terms of either price or output. This function can then be differentiated and set equal to zero to obtain a maximum. The second-order conditions should also be examined in this case to verify that the profit is indeed maximized, rather than minimized. This approach can be used in Problem 8.1 in particular, since that question specifically requires a

profit function to be obtained. The profit function approach is also used in Chapter 10 to deal with more complex demand functions involving other elements in the marketing mix.

Summary

- 1 Market structure describes the different conditions in markets that affect the way in which prices and outputs are determined.
- 2 The four main types of market structure are perfect competition, monopoly, monopolistic competition and oligopoly.
- 3 Market structure, conduct, performance and technology are all interdependent.
- 4 The determination of price and output can be examined graphically or algebraically.
- 5 In any type of market the profit-maximizing output is always given by the condition $MC = MR$.
- 6 Firms can only make supernormal profit in the long run if there are barriers to entry and exit.
- 7 Barriers can be either structural or strategic.
- 8 When comparing the performance of markets the key variables to examine are price, output, profits and efficiency (both productive and allocative).
- 9 Allocative efficiency is concerned with the optimality of resource allocation from the point of view of the economy as a whole, considering the effects on both consumer and producer. This has important implications for government policy.
- 10 Oligopoly is the most complicated type of market structure to analyse, since the strategic decisions of firms are interdependent.
- 11 Oligopoly is in practice the most important type of market structure, since the majority of most countries' output is produced in this type of market structure. This is especially true if we consider that many markets that appear to feature monopolistic competition are, in reality, limited intersecting oligopolies that are differentiated in terms of product and spatial characteristics. Restaurants are a good example.

Review questions

- 1 Why is perfect competition normally regarded as being 'better' than monopoly?
- 2 In what ways may perfect competition not be 'perfect'?
- 3 Explain the meaning and significance of limited intersecting oligopolies.
- 4 What is meant by monopoly power? What factors determine the extent of this power?
- 5 Explain why OPEC has been one of the most successful cartels in recent decades. What factors have limited this success?

- 6 Explain what is meant by the kinked demand curve. What shortcomings does this approach have in the analysis of oligopoly?

Problems

8.1 An apartment block has seventy units of accommodation. It is estimated that it is possible to let them all if the rent is \$2,000 per month, and for each \$100 per month the rent is increased there would be one unit vacant. LG, the manager of the block, finds that a vacant unit costs \$100 per month to maintain whereas an occupied one costs \$300.

- If profit from the lettings is measured as revenue minus maintenance costs, find an expression for profit in terms of the number of units let.
- What rent should LG charge to maximize profit?
- KA, a contractor, offers to be responsible for the maintenance of the entire block at a rate of \$150 per unit, whether the units are occupied or not. Would it be more profitable for LG to employ KA?

8.2 XL Corp has estimated its demand and cost functions to be as follows:

$$P = 60 - 0.2Q$$

$$C = 200 + 4Q + 1.2Q^2$$

where Q is in units, P is in \$ and C is in \$.

- Calculate the profit-maximizing price and output.
- Calculate the size of the profit.
- Calculate the price elasticity of demand at the above price.
- If there is a \$14 tax placed on the good, so that the producer has to pay the government \$14 for every unit sold, calculate the new profit-maximizing price and output.
- What would happen to profit if the firm tried to pass on all the tax to the consumer in the form of a higher price?
- If fixed costs rise by \$200 how would this affect the firm's situation?

8.3 Lizzie's Lingerie started selling robes for \$36, adding a 50 per cent mark-up on cost. Costs were estimated at \$24 each: the \$10 purchase price of each robe, plus \$6 in allocated variable overhead costs, plus an allocated fixed overhead charge of \$8. Customer response was such that when Lizzie's raised prices from \$36 to \$39 per robe, sales fell from 54 to 46 robes per week.

- Estimate the optimal (profit-maximizing) pricing strategy assuming a linear demand curve.
- Estimate the optimal pricing strategy assuming a power demand curve.
- Explain why there is a difference between the above two strategies.
- Estimate the size of the profit at both prices, assuming a power demand curve.

- e. Estimate the optimal price if the cost of buying the robes rises from \$10 to \$11, assuming a power demand curve.

8.4 Crystal Ball Corp. has estimated its demand and cost functions as follows:

$$Q = 80 - 5P$$

$$C = 30 + 2Q + 0.5Q^2$$

where P is in \$, Q is in thousands of units and C is in \$,000.

- a. Calculate the profit-maximizing price and output.
- b. Calculate the size of the above profit.
- c. Calculate the price elasticity of demand at the above output; is demand elastic or inelastic here? What should it be?
- d. Calculate the marginal cost at the above output.
- e. If unit costs rise by \$2 at all levels of output and the firm raises its price by the same amount, what profit is made?
- f. What is the profit-maximizing strategy given the above rise in costs?
- g. How much profit is the firm forgoing by raising its price \$2?

Notes

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